TASNEEM MUSTAFA

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Education:

BSc in Control and Automation Engineering *Istanbul Technical University* *Sep. 2023 – present*

Technical Summary:

Ambitious Control and Automation Engineering student at Istanbul Technical University with a strong focus on autonomous systems and intelligent robotics. Experienced in perception, planning, and control through hands-on projects using Python, C, ROS, Gazebo, OpenCV, MATLAB/Simulink, and Linux-based systems. Motivated to design reliable autonomous solutions that bridge simulation and real-world hardware.

Professional Experience:

BAYKAR Stajyer Mühendis – Autopilot and Intelligent Systems Software *June 2026- present*

- *Selected for a competitive engineering internship focusing on autopilot development and intelligent flight control software stacks.*

ZENITH ROBOTAKSI TEKNOFEST TEAM *Nov.2025 – present*

- Founded and led a cross-functional team for the Ready Vehicle category, defining the roadmap from simulation to real-world autonomy.
- Designing a ROS-based modular autonomy stack, implementing EKF-based localization and sensor fusion (LiDAR & camera) and integrating deep-learning perception with local path planning.
- Establishing a Gazebo-based validation pipeline to test control logic, safety constraints, and system integration prior to hardware deployment.

XENOS ROBOTAKSI TEKNOFEST FINALIST *Oct.2024 – July 2025*

- Worked on the development of an autonomous vehicle system within a simulation-based environment. Contributed to perception and control-related components using ROS, applying control theory and kinematic reasoning to support navigation and motion behavior.
- Utilized simulation tools to test and validate system responses, focusing on modular design, real-time decision-making, and reliable integration between perception outputs and vehicle motion

NOVA MATLAB MINIDRONE FINALIST *Jan.2025 – May 2025*

- Developed an image-based navigation algorithm for an autonomous minidrone in simulation.
- Designed a robust line detection and tracking pipeline using computer vision techniques, including edge detection and region-based analysis, to generate navigation error signals. Integrated perception outputs with control logic to enable stable position correction without yaw adjustments

Undergraduate Researcher – ITU ROBOTICS LAB *Oct.2025 – Jan.2026*

- Conducting research on Deep Reinforcement Learning (DRL) for ongoing humanoid robot project.
- Trained and deployed RL agents for pick-and-place tasks, optimizing trajectory planning and grasping stability NVIDIA Isaac Sim.

WINGS S_IHA Teknofest team *Feb.2025 – June 2025*

- Developed ROS-based mission simulations in Gazebo and contributed to real-time object detection and tracking for autonomous UAV tasks.

FEZA CanSat Competition Team Aug.2024 – jan.2025

- Built mission-level FSW with state management, real-time telemetry, and sensor fusion on a Teensy-based system.

Ball Balancing Robot *Sep. 2025 – Oct.2025*

- Developed a real-time computer vision pipeline using OpenCV and Python to estimate object position.
- Designed serial communication between PC and STM32 and implemented PID-controlled servo stabilization.
- Focused on closed-loop system integration between embedded firmware and vision-based perception.

TECHNICAL SKILLS

Programming: Python, C, MATLAB, Embedded Systems: STM32, Teensy, Arduino, Alakart
Robotics & Simulation: ROS1/ROS2, Gazebo, NVIDIA Isaac Sim, MATLAB/Simulink
Operating Systems: Linux, Domains: Computer Vision, Image Processing, Localization, Control Theory, Deep Learning / DRL

Courses:

SELF DRIVING CAR SPECIALIZATION University of Toronto

Languages: **English | Arabic| Turkish | French**